Analysis of Rayleigh Number Sensitivity in Echo State Network Forecasting of Lorenz Systems

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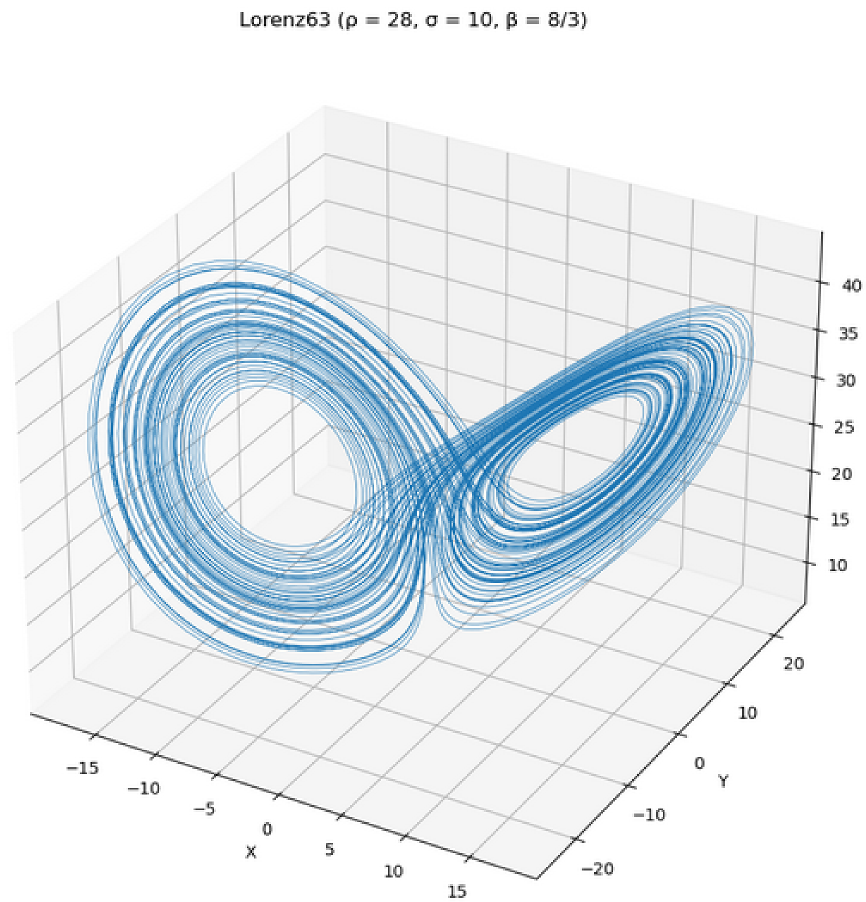

FIG. 1. Lorenz63 system trajectory at Rayleigh number $\rho = 28$.

Abstract

We train an echo state network (ESN) to predict the Lorenz-63 system. We then evaluate the error sensitivity of this network across different values of the Lorenz system parameter called the Rayleigh number. We observe that the error rises by several orders of magnitude as we vary our Rayleigh number. This demonstrates the limitations of echo state networks in real-world dynamical environments where the Rayleigh number is not constant, posing a careful consideration for climate scientists looking for computationally efficient alternatives in climate modeling of chaotic systems.

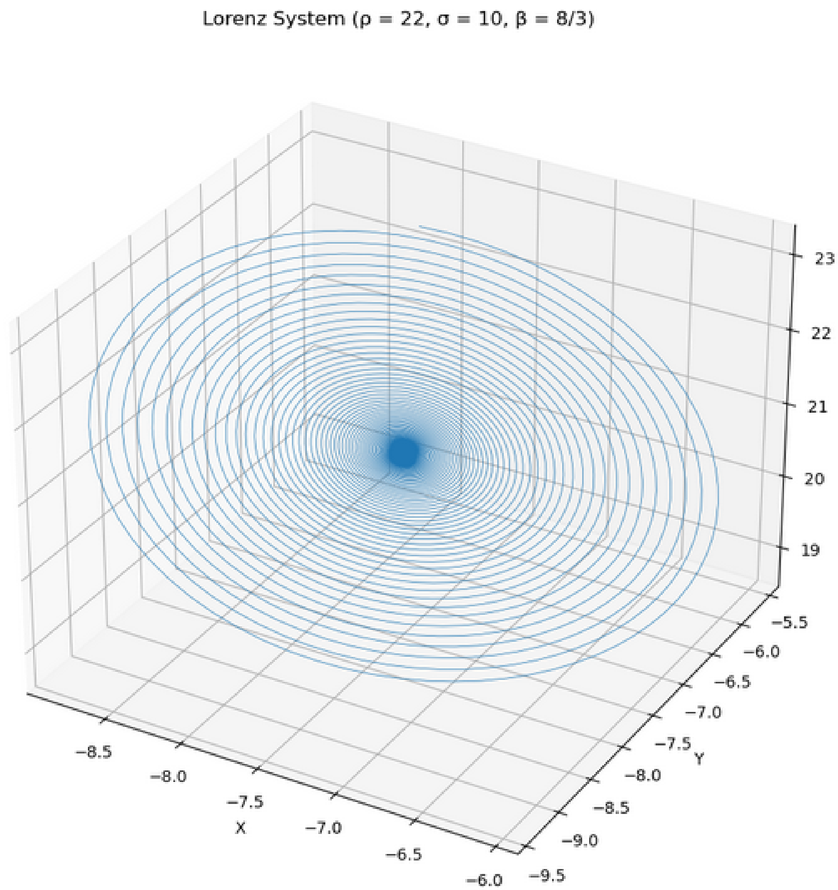

FIG. 2. Lorenz system trajectory at Rayleigh number $\rho = 22$. This system is not chaotic.

I. INTRODUCTION

The Lorenz system is a set of three differential equations that exhibit highly chaotic behavior:

$$\dot{x} = \sigma(y - x), \tag{1}$$

$$\dot{y} = x(\rho - z) - y, \tag{2}$$

$$\dot{z} = xy - \beta z. \tag{3}$$

where $(x, y, z) \in \mathbb{R}^3$, often with a physical interpretation of space, and $t \in \mathbb{R}^+$, often with a physical interpretation of time. σ, ρ, β are parameters of the Lorenz system, with ρ influencing the geometry of the system (bifurcations, curvature, "butterfly" shape, etc.). The iconic Lorenz system, named Lorenz-63, has the parameters $(\rho, \sigma, \beta) = (28, 10, 8/3)$ [1]. This system is known for its highly chaotic behavior (see Fig. 1).

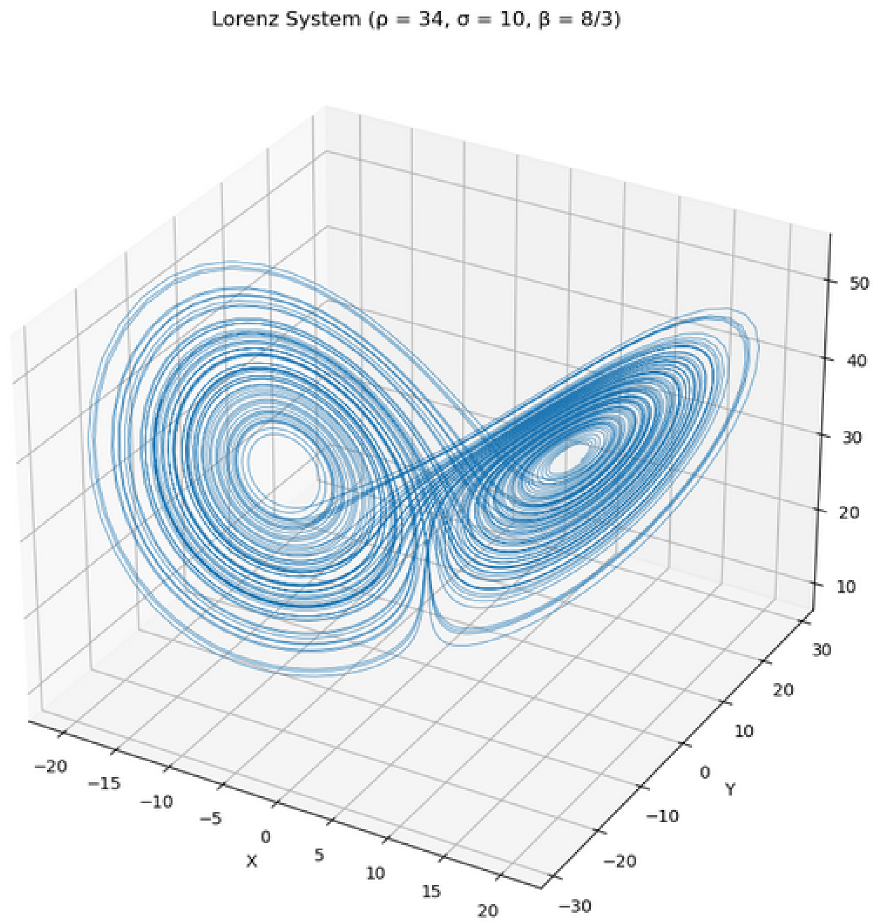

FIG. 3. Lorenz system trajectory at Rayleigh number $\rho = 34$.

The Lorenz system naturally emerges as a model for many physical phenomena. Prominently, we can use the Lorenz system to represent a simplified version of Rayleigh-Bénard convection, the physical behavior describing how air moves along a temperature gradient. This physical behavior can be incorporated into climate models for more accurate predictions. However, climate models are often constrained by their computational demand. For example, if we simulate the heat convection of our Earth using a box model, as we increase the number of boxes, our accuracy goes up for longer time-window predictions, but this also increases computational complexity (typically by polynomial time). Therefore, we need ways of simplifying computations. One method is to use recurrent neural networks, specifically echo state networks (ESNs), to predict our Lorenz system [2]. However, robustness against real world conditions in which ρ differs from the training dataset (such as Lorenz63) is underexplored. Demonstrating acceptable error margins across ρ variation would bolster the viability of echo-state network applications in climate science. For Lorenz systems with different Rayleigh numbers see Fig. 2 and Fig. 3.

II. THEORY

A. Lorenz System

In order to provide training data and data on Lorenz systems with different ρ values to cross reference our ESN predictions with, we first have to solve for our system. No close-form solution is known, so we numerically evaluate our equations with a set of random initial conditions. The numerical evaluation uses the function scipy.integrate.odeint and time step $\Delta t = 0.01$.

B. Echo State Networks (ESNs)

ESNs are a specific type of recurrent neural network that uses reservoir computing. A reservoir computer assigns random fixed weights and connections to its nodes, and only trains the nodes connected to its output layer. This gives the network higher dimensionality while turning training into a linear regression problem. We tune our ESN by modifying hyperparameters: connectivity, leakage rate, and spectral radius. Leakage rate (lr) controls the rate at which our ESN forgets past information, spectral radius (sr) controls the ESNs memory capacity and "chaotic" behavior [3, 4].

III. METHODOLOGY

A. Choosing Hyperparameters

We have heuristically chosen our hyperparameters based on the resulting MSE of Lorenz63 predictions. Due to the high variation in mean square error (MSE) predictions based on hyperparameter selection, our results are likely suboptimal and can be greatly improved by performing a Bayesian optimization on our hyperparameters for least MSE; however, our findings still act as a proof of concept. We set $sr = 0.8, lr = 0.390, \text{input connectivity} = 0.10, \text{reservoir connectivity} = 0.30$, and our number of nodes =

400.

B. Training

We apply a warm up to our ESN training of 200 time steps to ensure our ESN has a memory that locks onto our Lorenz attractor (an attractor is the general area a chaotic system trends toward). Afterward we train our ESN for 17,800 time steps using data we numerically produced for our Lorenz63 system.

C. Evaluating Our Model

Once we have trained our ESN, we wipe our state memory, warm up our ESN on 500 time steps, and predict the next 100 time steps (these time steps were chosen for numerical accuracy in the integrator solution). We calculate the averaged mean square error (MSE) between 200 trials of the numerically produced Lorenz system and 200 trials of each of our predicted Lorenz systems. We numerically evaluate Lorenz systems at $\rho = 22, 24, 26, 28, 30, 32, 34$.

IV. RESULTS AND ANALYSIS

MSE for $\rho = 28$ was three orders of magnitude smaller than all other Lorenz systems (see Fig. 4). As preliminary results, these suggest ESNs trained on a particular Rayleigh number do not generalize across ρ values. However, these findings should be interpreted cautiously. It's possible that a Bayesian optimization could reduce error across ρ values by several orders of magnitude. A thorough exploration of hyper parameter space is necessary to confirm the lack of viability of ESN generalization across ρ values.

ρ	100-step MSE (± std)
22.0	1.32e+02 ± 1.36e-01
24.0	9.14e+01 ± 5.34e+00
26.0	7.22e+01 ± 7.27e+01
28.0	2.12e-01 ± 1.54e+00
30.0	8.35e+01 ± 6.61e+01
32.0	1.19e+02 ± 7.46e+01
34.0	1.53e+02 ± 7.78e+01

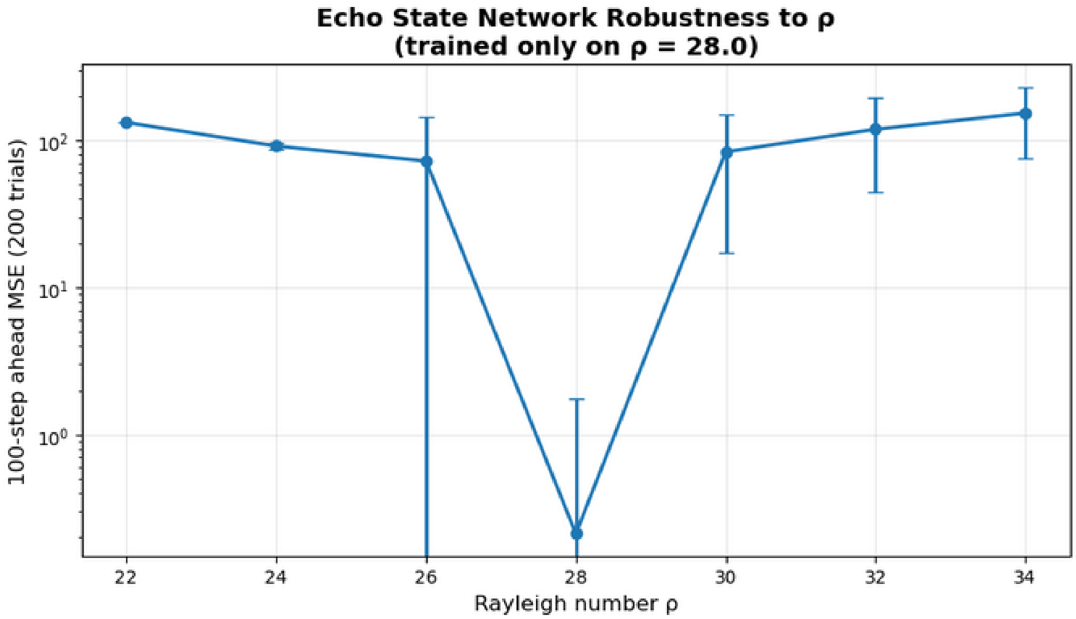

FIG. 4. MSE (log) scale against ρ.

V. CONCLUSION

While ESNs have demonstrated their ability to act as computationally efficient predictors for Lorenz systems, our data suggests Lorenz system trained ESNs generalize across different ρ values poorly. To verify this claim, we suggest a Bayesian optimization is done to search hyperparameter space; if minimization still results in poor generalization, this would further confirm that ESNs poorly generalize across ρ values. Furthermore, these findings would suggest climate scientists should be careful about employing ESNs in environments where ρ varies significantly and restrain ESN usage to environments with steady heat convection rates.

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[4] H. Jaeger. *The "Echo State" Approach to Analyzing and Training Recurrent Neural Networks.* GMD Technical Report 148, German National Research Center for Information Technology, 2001. URL: `https://www.ai.rug.nl/minds/uploads/EchoStatesTechRep.pdf`.